# Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.

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# Project Overview

Analyzing 3,900 purchases to understand spending patterns, segments, preferences, and subscription behavior.

### 3,900 Purchases

Extensive transactional data.

### Uncover Insights

Spending, segments, preferences.

### Guide Decisions

Strategic business growth.

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# Dataset Summary

Our dataset comprises 3,900 rows and 18 columns, capturing a comprehensive view of customer interactions.

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item, Category, Amount, Season, Size, Color)
- Shopping behavior (Discount, Promo, Previous Purchases, Frequency, Review, Shipping)

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# Exploratory Data Analysis (Python)

Data preparation and cleaning were crucial steps in Python.

01

## Data Loading & Initial Exploration

Imported dataset, checked structure and summary statistics.

02

## Missing Data Handling

Imputed 37 missing 'Review Rating' values using median per category.

03

## Column Standardization

Renamed columns to snake case for readability.

04

## Feature Engineering

Created 'age_group' and 'purchase_frequency_days'.

05

## Data Consistency & Integration

Dropped redundant 'promo_code_used', loaded to PostgreSQL.

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# Key Business Questions (SQL Analysis)

Structured analysis in PostgreSQL addressed critical business inquiries.

## Revenue by Gender

Comparing total revenue from male vs. female customers.

## High-Spending Discount Users

Customers using discounts but spending above average.

## Top 5 Products by Rating

Products with the highest average review ratings.

## Shipping Type Comparison

Average purchase amounts for Standard vs. Express shipping.

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# Subscription & Discount Insights

## Subscribers vs. Non-Subscribers

Average spend and total revenue across subscription statuses.

## Discount-Dependent Products

Top 5 products with the highest percentage of discounted purchases.

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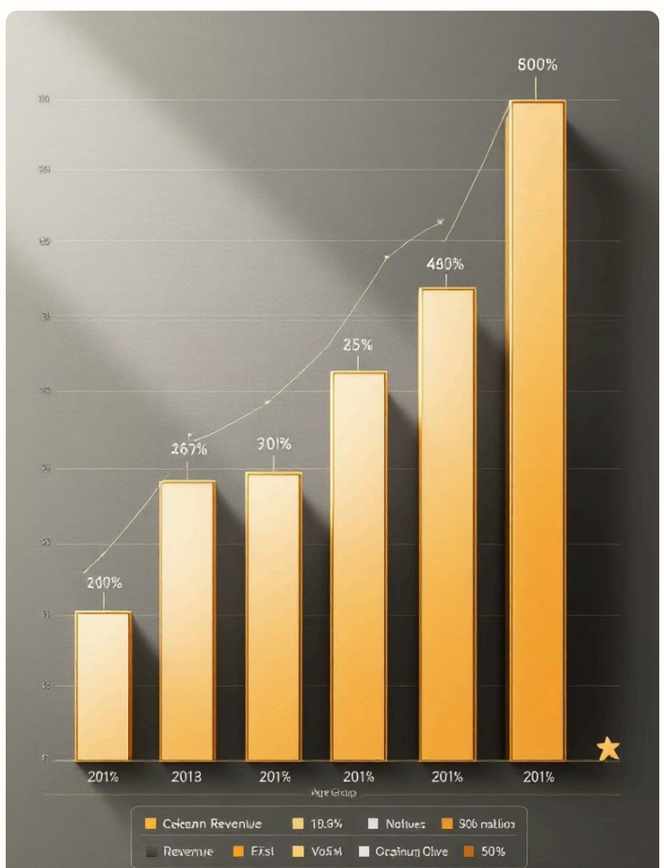
# Customer Segmentation & Age Group Revenue

## Customer Segmentation

Classified customers into New, Returning, and Loyal segments.




## Revenue by Age Group

Total revenue contribution from each age group.

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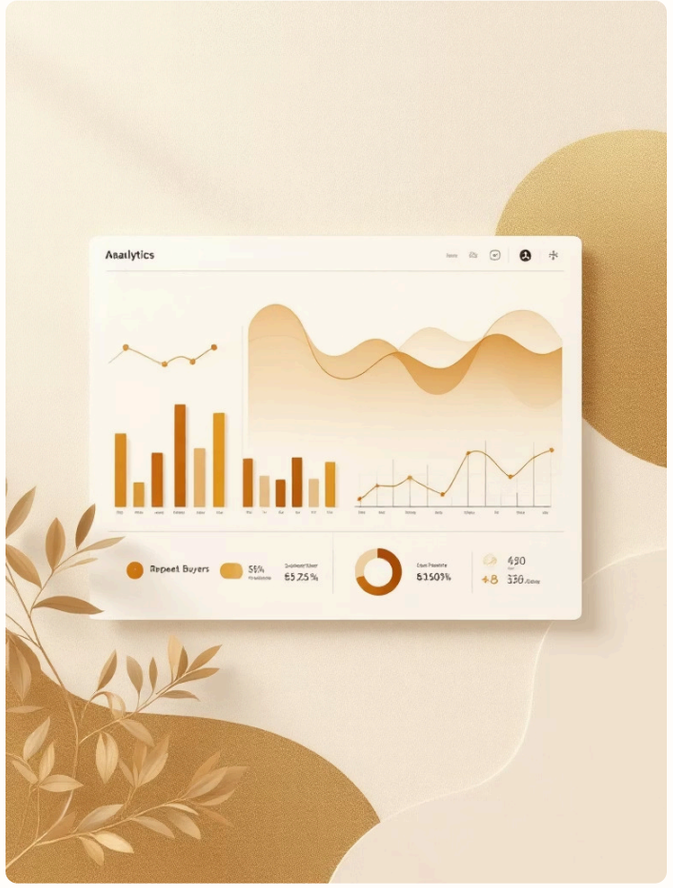
# Product & Loyalty Deep Dive

## Top 3 Products per Category

Most purchased products within each category identified.




## Repeat Buyers & Subscriptions

Examined if customers with >5 purchases are more likely to subscribe.

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# Interactive Dashboard in Power BI

Visualizing insights for easy comprehension and decision-making.

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# Business Recommendations

### Boost Subscriptions

Promote exclusive benefits.

### Customer Loyalty Programs

Reward repeat buyers.

### Review Discount Policy

Balance sales with margin control.

### Product Positioning

Highlight top-rated and best-selling products.

### Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

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